

Convection-diffusion problems

Andrea Cangiani

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REFERENCES

Ern–Guermond *Theory and Practice of Finite Elements*, (Chapter 3) Springer, 2004.

Johnson, *Numerical solution of partial differential equations by the finite element method*, Dover, 2009

General elliptic operator in $\mathbb{R}^d$

Consider the second-order differential operator

$$\mathcal{L}u = -\nabla \cdot (A \nabla u) + \beta \cdot \nabla u + \mu u,$$

with $A = (a_{ij})_{ij}$, $\beta = (\beta_i)_i$. That is, diffusion + advection + reaction.

Assume $A(x)$ is positive definite a.e. in Ω . Then the quadratic term

$$a_0(v, v) = \int_{\Omega} A(x) \nabla v \cdot \nabla v \, dx$$

is coercive:

$$a_0(v, v) \geq \alpha_0 \|v\|_{H_0^1(\Omega)}^2 \quad \forall v \in H_0^1(\Omega).$$

What is new?

We already know how to discretize diffusion.

The new issues are:

- ▶ discretization of advection / transport,
- ▶ interplay of advection and diffusion.

Assume homogeneous Dirichlet boundary conditions, so that

$$V = H_0^1(\Omega),$$

and let $V_h^k \subset V$ be a standard FEM space.

Weak form: find $u_h \in V_h^k$ such that

$$\int_{\Omega} A \nabla u_h \cdot \nabla v_h + \int_{\Omega} (\beta \cdot \nabla u_h) v_h + \int_{\Omega} \mu u_h v_h = \int_{\Omega} f v_h \quad \forall v_h \in V_h^k.$$

Coercivity of the full bilinear form

The bilinear form is

$$a(u_h, v_h) = \int_{\Omega} A \nabla u_h \cdot \nabla v_h + \int_{\Omega} (\beta \cdot \nabla u_h) v_h + \int_{\Omega} \mu u_h v_h.$$

A sufficient coercivity condition is

$$\mu - \frac{1}{2} \nabla \cdot \beta \geq 0 \quad \text{a.e. in } \Omega.$$

In particular, if $\mu = 0$ and $\nabla \cdot \beta = 0$, then

$$a(u_h, u_h) = \int_{\Omega} A \nabla u_h \cdot \nabla u_h + \frac{1}{2} \int_{\Omega} \beta \cdot \nabla (u_h^2).$$

After integration by parts, the transport contribution does not destroy coercivity, so coercivity of the full bilinear form relies only on positivity of the diffusion term.

A priori estimate and advection domination

Under coercivity we have the usual estimate

$$\|u - u_h\|_{H^1(\Omega)} \leq \frac{\gamma}{\alpha_0} \inf_{v_h \in V_h^k} \|u - v_h\|_{H^1(\Omega)},$$

where roughly

$$\gamma \sim \|A\|_{L^\infty} + \|\beta\|_{L^\infty} + \|\mu\|_{L^\infty}.$$

Hence **if $\alpha_0 \ll 1$, the error may be potentially large...**

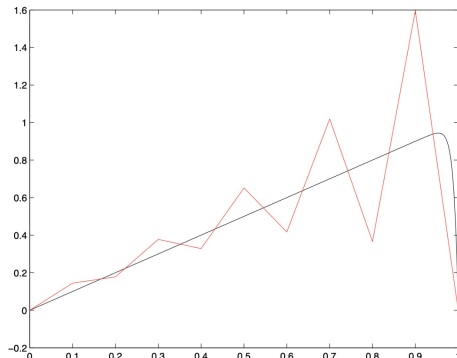
Advection-dominated problem: 1D example

Example in 1D:

$$-\varepsilon u'' + u' = 1 \quad \text{in } (0, 1), \quad u(0) = u(1) = 0,$$

with $\varepsilon = 10^{-2}$. This is a (mildly) advection-dominated problem:

- ▶ the true solution behaves like the solution of a pure transport problem outside $[1 - \varepsilon, 1]$;
- ▶ standard linear FEM produces **unphysical oscillations over the whole domain**.



Pure transport (step 9): streamline diffusion method

Also called:

- ▶ Petrov–Galerkin,
- ▶ streamline upwind.

Model problem:

$$\mathcal{L}u := \beta \cdot \nabla u + \mu u = f \quad \text{in } \Omega,$$

with inflow boundary condition

$$u|_{\partial\Omega^-} = g.$$

Here

$$\partial\Omega^- := \{x \in \partial\Omega : \beta(x) \cdot n(x) < 0\}, \quad \partial\Omega^+ := \partial\Omega \setminus \partial\Omega^-.$$

Natural graph space

For pure transport the right ambient space is the **graph space**

$$W := \{w \in L^2(\Omega) : \beta \cdot \nabla w \in L^2(\Omega)\}.$$

Hilbert with norm

$$\|w\|_W^2 = \|w\|_{L^2(\Omega)}^2 + \|\beta \cdot \nabla w\|_{L^2(\Omega)}^2.$$

Then the natural trial space is

$$V_g := \{w \in W : w|_{\partial\Omega^-} = g\}.$$

If

$$\mu - \frac{1}{2} \nabla \cdot \beta \geq \mu_0 > 0 \quad \text{a.e. in } \Omega,$$

then

$$\mathcal{L} : W \rightarrow L^2(\Omega)$$

is an isomorphism.

Why not standard FEM directly?

To apply FEM we would like to multiply by a test function and integrate.
A standard choice would be once again with continuous FEM space

$$V_h \subset H^1(\Omega).$$

The boundary condition is only needed on $\partial\Omega^-$

A useful idea:

standard Galerkin + weak imposition of the inflow boundary condition.

Weak imposition of inflow boundary conditions

Find $u_h \in V_h^k$ such that

$$a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in V_h^k,$$

where

$$a(u_h, v_h) := (\beta \cdot \nabla u_h + \mu u_h, v_h)_\Omega - (u_h, v_h \beta \cdot n)_{\partial\Omega^-},$$

$$\ell(v_h) := (f, v_h)_\Omega - (g, v_h \beta \cdot n)_{\partial\Omega^-}.$$

This is satisfied by the exact solution, hence

$$a(u - u_h, v_h) = 0 \quad \forall v_h \in V_h^k,$$

that is, Galerkin orthogonality.

Coercivity of the transport bilinear form

For $v \in H^1(\Omega)$,

$$a(v, v) = (\beta \cdot \nabla v, v)_{\Omega} + (\mu v, v)_{\Omega} - (v, v \beta \cdot n)_{\partial\Omega}.$$

By Green's formula,

$$(\beta \cdot \nabla v, v)_{\Omega} = \frac{1}{2} \int_{\Omega} \beta \cdot \nabla (v^2) = -\frac{1}{2} \int_{\Omega} (\nabla \cdot \beta) v^2 + \frac{1}{2} \int_{\partial\Omega} \beta \cdot n v^2.$$

Therefore

$$a(v, v) = \int_{\Omega} \left(\mu - \frac{1}{2} \nabla \cdot \beta \right) v^2 + \int_{\partial\Omega} |\beta \cdot n| v^2$$

Boundary seminorm and coercivity estimate

So,

$$a(v, v) = \int_{\Omega} \left(\mu - \frac{1}{2} \nabla \cdot \beta \right) v^2 + \frac{1}{2} \int_{\partial\Omega} |\beta \cdot n| v^2.$$

Define

$$|v|_{\beta, \partial\Omega}^2 := \int_{\partial\Omega} |\beta \cdot n| v^2.$$

Then, if

$$\mu - \frac{1}{2} \nabla \cdot \beta \geq \mu_0 > 0,$$

we obtain

$$a(v, v) \geq \mu_0 \|v\|_{L^2(\Omega)}^2 + \frac{1}{2} |v|_{\beta, \partial\Omega}^2 \quad \forall v \in H^1(\Omega).$$

So **coercivity holds**, but only **in an L^2 -type norm**.

Error estimate from the notes

The method is consistent and stable, and one obtains the estimate quoted in the notes¹:

$$\|u - u_h\|_{L^2(\Omega)} + |u - u_h|_{\beta, \partial\Omega} \leq Ch^k \|u\|_{H^{k+1}(\Omega)} \quad \text{if } u \in H^{k+1}(\Omega).$$

Main limitations highlighted in the notes:

- ▶ suboptimal in the L^2 norm,
- ▶ requires smooth exact solution u , while solutions may be discontinuous along characteristics,
- ▶ In that regime, standard Galerkin lacks the correct directional control and may generate oscillations.

This is the motivation for adding stabilization in the streamline direction:

streamline diffusion / SUPG.

¹[Johnson, Numerical solution of partial differential equations by the finite element method, Dover, 2009]

SUPG / streamline diffusion for pure advection

recall that coercivity was based on L^2 -norm control only. We need to add some diffusion...

Petrov-Galerkin approach: test with

$$v_h + \delta \beta \cdot \nabla v_h \quad \text{for } v_h \in V_h$$

where typically $\delta \sim h$ (or cellwise $\delta_K \sim h_K$).

This gives: find $u_h \in V_h^k$ such that

$$a_h(u_h, v_h) = \ell_h(v_h) \quad \forall v_h \in V_h^k,$$

with

$$(\beta \cdot \nabla u_h + \mu u_h, v_h + \delta \beta \cdot \nabla v_h)_\Omega - (u_h, \beta \cdot n v_h)_{\partial\Omega^-} = (f, v_h + \delta \beta \cdot \nabla v_h)_\Omega - (g, \beta \cdot n v_h)_{\partial\Omega^-}$$

Note: as $v_h + \delta \beta \cdot \nabla v_h \notin V_h$, test and trial live in different spaces.

Streamline-diffusion norm and stability

The added term

$$(\beta \cdot \nabla u_h, \delta \beta \cdot \nabla v_h)_\Omega$$

acts as **artificial diffusion in the streamline direction**. Yields control of the directional derivative $\beta \cdot \nabla v \dots$

We can define the (stronger) **streamline diffusion norm**

$$\|v\|_\beta := \left(h \|\beta \cdot \nabla v\|_{L^2(\Omega)}^2 + \|v\|_{L^2(\Omega)}^2 + \frac{1+h}{2} |v|_{\beta, \partial\Omega}^2 \right)^{1/2},$$

where

$$|v|_{\beta, \partial\Omega}^2 := \int_{\partial\Omega} |\beta \cdot n| v^2.$$

The method is still consistent, so Galerkin orthogonality still holds, but now there is extra stability in the streamline derivative.

SUPG a priori analysis

Lemma (stability)

For all $v \in H^1(\Omega)$,

$$a_h(v, v) \geq \|v\|_\beta^2.$$

Lemma (error estimate)

If $u \in H^{k+1}(\Omega)$, then there exists $C > 0$ independent from h :

$$\|u - u_h\|_\beta \leq Ch^{k+1/2} \|u\|_{H^{k+1}(\Omega)}, \quad \text{if } u \in H^{k+1}(\Omega).$$

Main message:

- ▶ standard Galerkin gives only L^2 -type control,
- ▶ SUPG adds stabilization in the streamline direction,
- ▶ the method remains consistent,
- ▶ oscillations are damped without switching to a fully discontinuous method (next lecture!).